

The University of Nottingham

SCHOOL OF Computer Science

A LEVEL THREE MODULE, SPRING SEMESTER 2017-2018

Computer Vision

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Out of FOUR Questions

Each question carries 20 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: None

INFORMATION FOR INVIGILATORS: None

Marking Schemes:

K - Knowledge
C - Comprehension
A - Application

1. The following questions are pertaining to Features and Image Segmentation.

- a. Patch based features are widely used in Computer Vision to describe certain characteristics of an image patch. To generate patch based features, it generally involves extracting some basic feature for each pixel and summarising the basic features over all the pixels in the patch.
- i. Describe the basic features that are extracted at the pixel level in Local Binary Pattern (LBP) and Histogram of Oriented Gradients (HOG), respectively. [3 Marks]
 - ii. Histogram is often used for summarising basic features over all the pixels in an image patch. Are such summarised features robust to rotation and translation of the image? Explain. [2 Marks]
 - iii. In both LBP and HOG methods, histograms of basic features are not computed over the entire image but instead in sub-windows. How are the histograms in sub-windows used to form final features of the image? Why is that strategy adopted? Explain. [2 Marks]

Answers:

- i. (K). For LBP, for each pixel p , create an 8-bit number $b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8$, where $b_i = 0$ if neighbour i has value less than or equal to p 's value and 1 otherwise. (1 mark) The binary string is then usually converted to an integer. (1 mark)

For HoG, the image gradient is calculated for each pixel in terms of its magnitude and orientation. (1 mark)
 - ii. (K+C). The histogram discards the spatial information of image pixels and therefore is robust to translations and rotations (1 mark), if they are smaller than spatial or orientation bin size (1 mark). [if the answer only addresses either translation or rotation, then award half of the full marks.]
 - iii. (K+C). The histograms of pixel level features in sub-windows are concatenated to form a final feature vector (1 mark). This is to capture the overall space arrangement of patches in an image (1 mark).
- b. An image is shown in Figure 1. You are asked to design an approach to segment the image into different regions. You need to make choices on a feature representation for each of the pixels in the image and a segmentation method to assign each of the pixels into its desired region. Three types of feature representation are available for you to choose, including histogram of intensity, LBP, and HOG. In order to obtain the feature for each of the pixels, it is computed over a small window that is centred on the pixel. The two segmentation methods that you can select are K-means and Gaussian Mixture Models (GMM).



Figure 1

- i. The task is to segment the image into three regions representing the person, the ground covered by the snow, and the sky. Which of the three features will you choose? You should evaluate each of them and justify your decision. [6 Marks]
- ii. Compare and contrast the principles and algorithmic procedures of k-means and GMM for image segmentation tasks. [5 marks]
- iii. Assume that you will use the histogram of intensity to represent each of the pixels in the image. Out of k-means and GMM, which one will you choose to perform the segmentation task in (i)? Explain. [2 Marks]

Answers:

- i. (C+A). The question does not have a unique answer. It is designed to test the understanding of the features and their suitability for image segmentation. The key components in an answer should cover what/how each feature would capture for different regions and whether they may or may not be discriminative. 2 marks for each feature type (1 mark for strength, 1 mark for weakness). A model solution could be as follows.

For histogram of intensity, it may differentiate between the person and the background, but may not work for snow and sky (2 marks). There seems to be apparent difference in texture among three regions, i.e., for the person it contains much smoother texture (if there's any at all), the snow has grainy texture, and the sky (with the cloud) has larger texture which may or may not be captured depending on the size of the window. (2 marks). HOG features are more prominent at the boundary of the person but not so within a region. If it is used with k-means or GMM, it probably is not that useful for region clustering. (2 marks)

- ii. (K). 1 mark for any of the following pair of points. Any five pairs of points are acceptable.

- | | |
|---|--|
| <ul style="list-style-type: none"> • K-means <ul style="list-style-type: none"> • Non-probabilistic model • Doesn't deal with uncertainty, noise, etc. • Iterative • Start with initial guess of mean values of the clusters • Assign each pixel to only one of the cluster • Update the mean of each cluster using only the pixels that are assigned to that particular cluster • Segmentation corresponds to the final assignment of each pixels | <ul style="list-style-type: none"> • GMM <ul style="list-style-type: none"> • Probabilistic model • Deal with uncertainty, noise • Iterative • Start with initial guess of the mean, covariance, weight of each of the Gaussian component • Calculate the probability that each pixel belongs to each of the Gaussian component • All pixels contribute to the computation/update of the mean, covariance, weight of each of Gaussian • Segmentation corresponds to the largest probability that a pixel belongs to a particular Gaussian |
|---|--|

- iii. (K+C). As the pixels in the snow and sky area may be very similar in terms of their histograms of intensity (1 mark), GMM may be preferred since it takes into the uncertainty of the features whilst k-means does not (1 mark). [However, any other reasonable answer is also acceptable]

2. The following questions are pertaining to Motion, Optical Flow, and Tracking.

- a. To evaluate the motion of moving objects, in both Horn-Schunk and Lucas-Kanade methods, the concept of optical flow is used and it must satisfy two constraints, i.e., brightness constancy and smoothness.

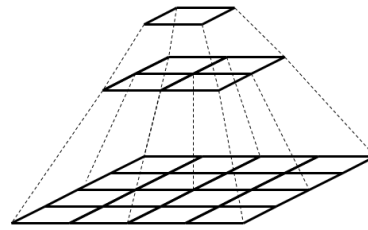
- | | |
|---|------------------|
| <p>i. Explain how smoothness constraint is expressed in Horn-Schunk and Lucas-Kanade, respectively.</p> | <p>[2 marks]</p> |
| <p>ii. Describe the concept of image pyramid and explain how it is used with Lucas-Kanade to overcome the issue caused by large motions.</p> | <p>[6 marks]</p> |
| <p>iii. Motion field is not necessarily the same as optical flow. Give an example and explain how the motion field differs from the optical flow in your example.</p> | <p>[2 marks]</p> |

Answers:

- i. (K). Lucas-Kanade assumes that all pixels in a local window have the same (u,v) as the pixel of interest (1 mark). Horn-Schunck assumes that u and v are equal to the average of their neighbouring values (1 mark)

ii. (K+C).

- A series of images
- Each is smaller than the last (usually half the size)
- Each pixel in a smaller image corresponds to a number of pixels (usually 4) in the larger one
- Images are often blurred before being reduced



- Images further up the pyramid are lower resolution, but can be processed faster

(2.5 marks)

- Pyramids allow us to use optical flow for large motion
- The motion (in pixels) is halved at each level
- By going up levels we can make any motion as small as we please
- After n levels, the motion is reduced by $1/2^n$, so don't usually need very many

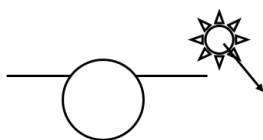
- Algorithm:
 - Start at the smallest level of the pyramid
 - Compute the optical flow
 - Use this to initialise the optical flow at the next level
 - Compute the optical flow at the new level
 - Repeat until you reach the last level

(2.5 marks)

Highlight that large motion can be reduced within 1 pixel in smaller level. (1 mark)

iii. (K+C)

- A Moving light



- The **image** changes so there is optical flow
- The **scene** objects do not move so there is no motion field

- A Rotating Sphere



- The **scene** object moves, so there is motion field
- The **image** does not change, so there is no optic flow

Either one of the above or similar situations. (1 mark for illustration, 1 mark for explanation)

- b. Suppose you are tracking the target that is highlighted by the bounding box in Figure 2. Particle filters use a set of samples to approximate the target probability density, where each sample is a particle, representing a hypothesised target state with some

probability being correct. CONDENSATION is the original algorithm that implements particle filters for tracking targets.

- i. Assume a particle hypothesising the target, is centred at where it is marked with an "O" that is next to the bounding box. Explain how you may measure the weightage of this particle. Is the weightage to be likely high or low according to your way of measurement? Elaborate.
[4 marks]
- ii. CONDENSATION requires maintain the same number of particles throughout the tracking process. How is it done? Elaborate your answer with the image in Figure 2.
[3 marks]
- iii. What advantage does particle filters have over optical flow methods, e.g., Lucas-Kanade, for tracking targets in real world? Explain with the situation in the image in Figure 2.
[3 marks]

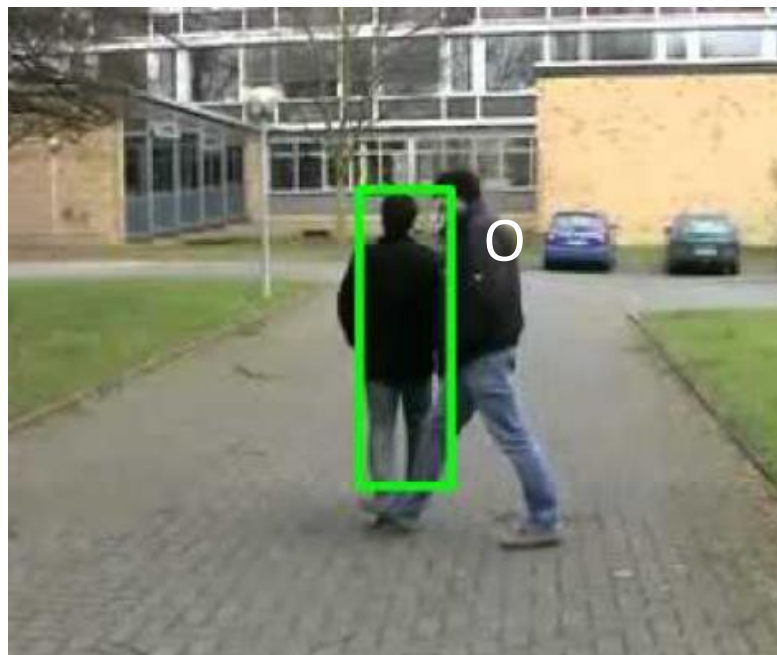


Figure 2

Answers:

- i. (K+A). Take the patch centred at "O" that is the same size of the bounding box (1 mark), compare the patch with the area within the bounding box and calculate its similarity or distance (1 mark). Note that one can use the patch directly or some kind of features summarised over the patch - same goes for the bounded area. The similar

the patch is compared to the bounded area, the higher the weightage of the particle (1 mark). As the patch centred at "O" in the image is quite similar to the bounded area, it is likely to be of high weightage (1 mark).

- ii. (K+A). After the particles being weighted according to the current measurement based on the image, the particles will be resampled with replacement (1 mark). The higher the weight a particle has, the more number of particles will be generated by this particular particle (1 mark). For example, the particle that was marked with an "O" is likely to be resampled more than once if it has higher weight than other particles. (1 mark)
- iii. (K+A). Lucas-Kanade assumes the probability distribution associated with target state is unimodal (1 mark), while the particle filters allow to track a target that has multimodal probability distribution (1 mark). In Figure 2, the target in the bounding box is very similar to the one being marked with an "O" in terms of their appearances and possibly the temporary movements. So potentially "O" can be a possible hypothesis of the target. (1 mark).

3. The following questions are pertaining to Stereo Vision.

- a. A typical stereo vision system usually consists of four stages, including camera calibration, image rectification, correspondence matching, and depth estimation.
 - i. Explain briefly the purposes of camera calibration and image rectification, respectively.

[2 marks]
 - ii. Correspondence matching is governed by epipolar geometry. Sketch and explain the main components of and the principle behind the epipolar geometry.

[4 marks]
 - iii. Patch based correlation method can be used to search the correspondences in the right image for the pixels in the left image in Figure 3. Discuss the effect of different patch window size (i.e., small vs. large) may have on searching the correspondences for the pixels in the regions marked with an "a" and a "b", respectively.

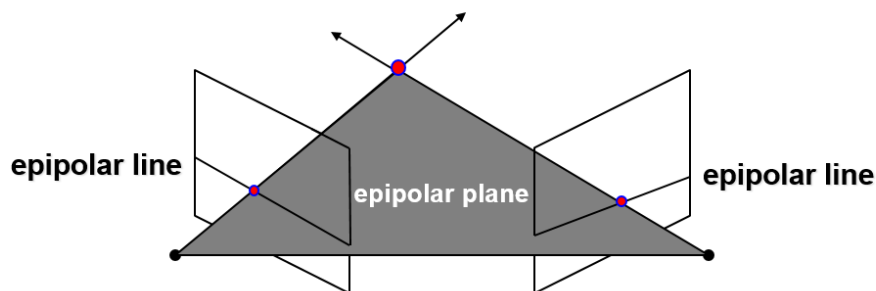
[4 marks]



Figure 3

Answers:

- i. (K). Camera calibration is to recover the intrinsic parameters of the camera(s) such as focal length, pixel size, principal point, and the cameras' relative pose including rotation, translation, and scale that transform left image coordinates onto right image (1 mark). Image rectification re-projects the images onto a common plane parallel to the line between the cameras, so that the epipolar lines are on the same scanline in the left and right image after the rectification (1 mark).
- ii. (K). Key components to check: image points (left and right), scene point, epipolar lines (left and right), epipoles (left and right), epipolar plane. 2 marks for the correct sketch and 2 marks for explanation of the relations among key components.



- Given the two optical centres and a point in one image, we can compute the epipolar plane and so the corresponding epipolar line in the other image
- iii. (C+A). Patches should be large enough to contain sufficient intensity variation, yet small enough to contain only pixels with about the same disparity. (1 mark). For a fixed window size, "a" contains higher intensity variation but less consistent disparity, compared to "b". (1 mark). Small window for "a" may be better than a larger one since it contains less inconsistent disparity but enough intensity variations. (1 mark). Large window for "b" is better since it will allow enough intensity variation to be covered and the disparity variation is still relatively small. (1 mark).

- b. Consider again the pair of rectified images in Figure 3 for stereo correspondence. Global methods represent dense correspondence between the two images as a continuous or discrete function defined over the coordinates of image pixels. For example, $d(x, y)$ is the disparity between the pixel at (x, y) in the left image and the pixel at $(x + d(x, y), y)$ in the right image. Matching correspondence is found to minimise a global cost function of image data and smoothness constraint.
- i. Explain briefly what aspects of the cost measurement the image data and smoothness constraint refers to, respectively. [2 marks]
 - ii. Given that pixel A matches A' and pixel B matches B', the ordering constraint requires that if A is to the left of B in the left image, A' must be to the left of B' in the right image. Ordering constraint is often useful but may be violated under certain circumstances. Sketch an example and explain how the ordering constraint is violated in your example. [4 marks]
 - iii. Dynamic programming can be used to find disparity along a particular scanline, where ordering constraint is normally assumed to be valid. An example of optimal path through grid is shown in Figure 4, where diagonal lines represent matched pixels between the left and right images, and horizontal or vertical lines represent unmatched pixels in either the left or right image, respectively. Using the style in Figure 4, sketch a possible path yielded by dynamic programming for the scanline highlighted as a dotted line in Figure 3. Your sketch should satisfy the ordering constraint, and reflect reasonably observable occlusions in left and right image respectively. Your sketch does not need to show the actual image width. Explain your sketch. [4 marks]

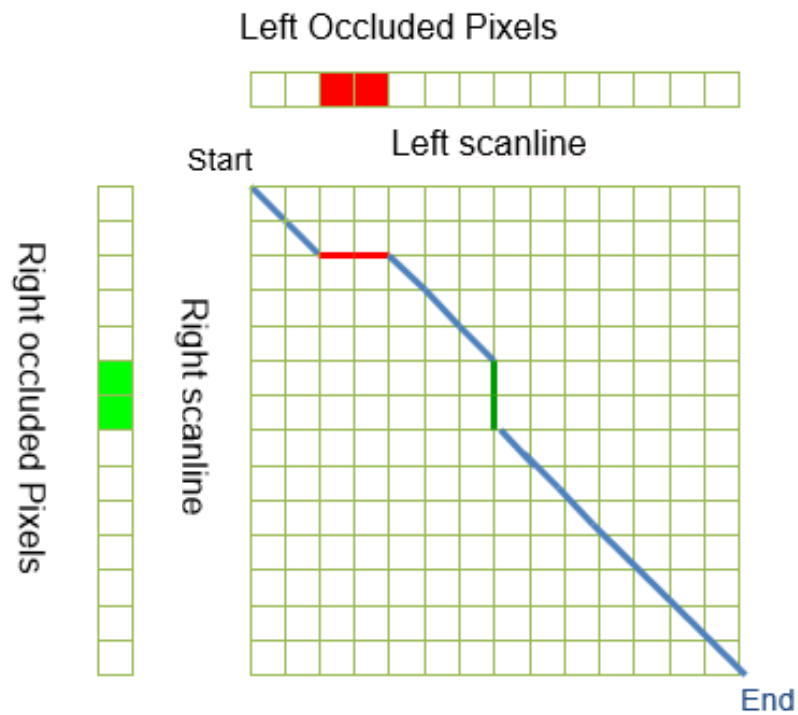
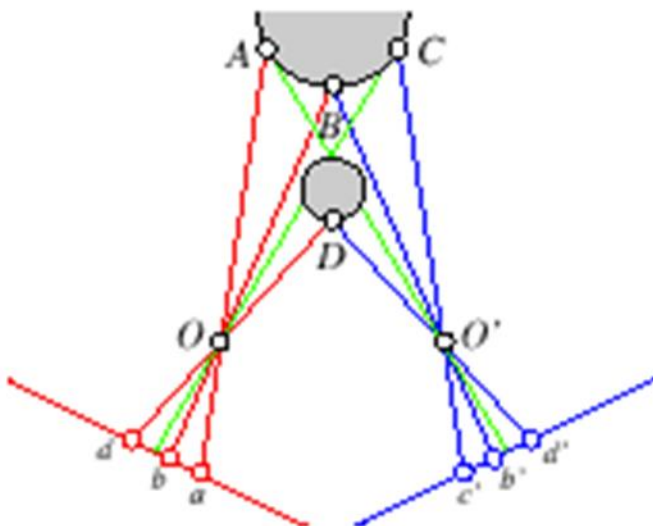


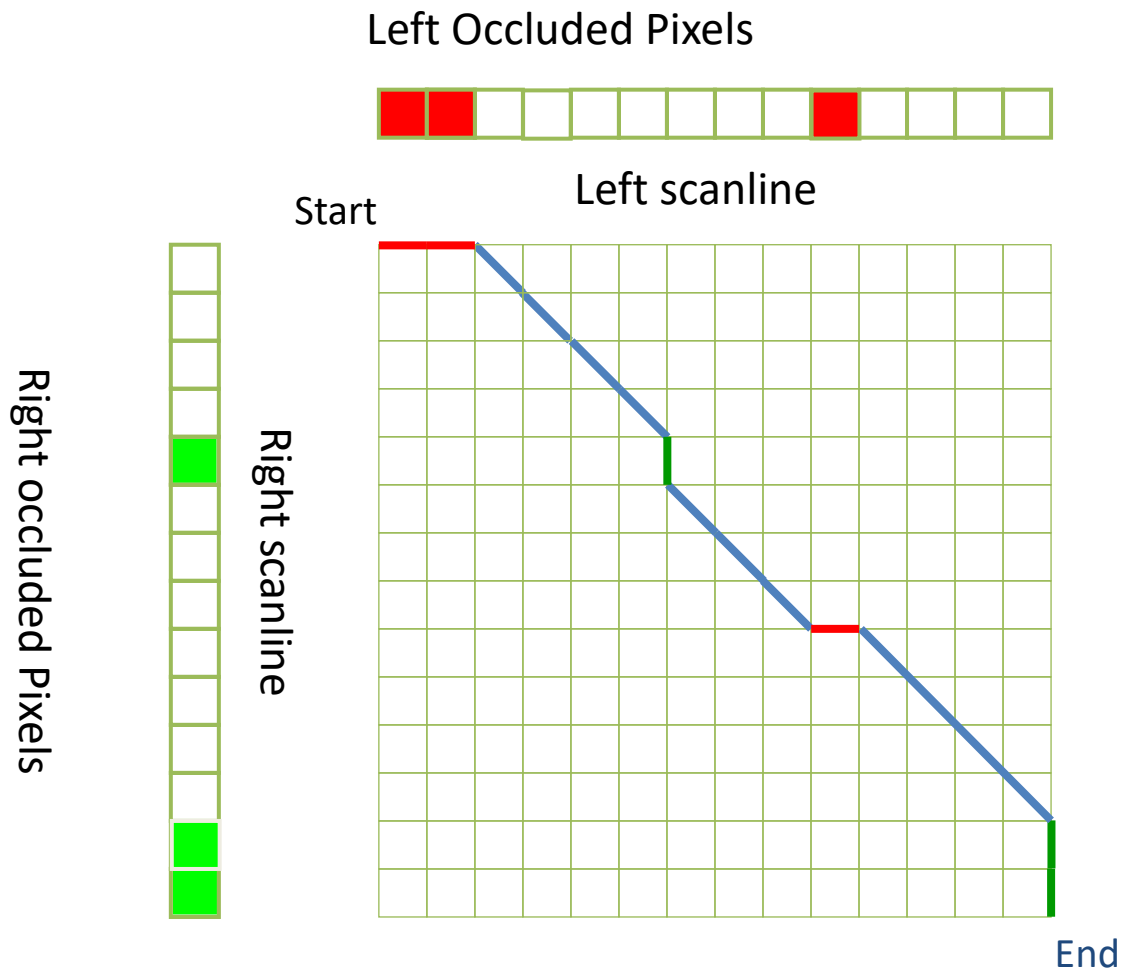
Figure 4

Answers:

- i. (K). Image data constraint refers to the measurement of the likelihood (or similar terms) that the right image is generated from the left image given the disparity map. (1 mark). Smoothness constraint requires that the disparities of pixels in a small neighbourhood should vary slowly (or similar to each other). (1 mark)
- ii. (K+C). 3 marks for the sketch. 1 mark for the explanation. Explanation should address the situation such as narrow objects.



- iii. (C+A).



1 mark for left occlusion. 1 mark for right occlusion. 2 mark for the explanation, e.g., the left occlusion refers to the black car on the left (1 mark), right occlusion refers to the edge of the building on the right (1 mark) etc.

4. The following questions are pertaining to Features, Object Recognition, and Machine Learning approaches.

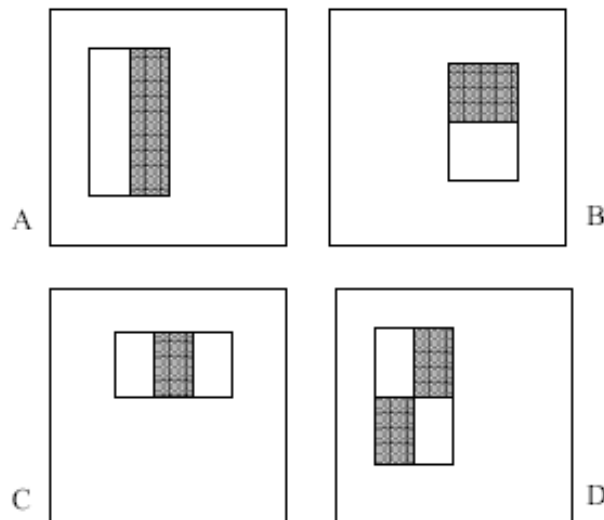
- a. Bag-of-Features is a feature representation technique that describes an image as a histogram of local image features. Each local image feature is analogous to a word in a document. Thus all different local image features would form a codebook, analogous to vocabulary.
- i. Describe the general process of constructing a codebook. [3 marks]
- ii. Usually the bag-of-features of an image is a sparse vector, i.e., only a few entries of the histogram of the codebook is non-zero. It is inefficient for the purpose of image retrieval in large image database. Inverted file technique can help alleviate the problem. Explain the concept of the inverted file and how it is used to improve the retrieval efficiency. [3 marks]

Answers:

- i. (K). The answer should cover the steps including feature extraction (patches, point feature based, etc. from large number of different images) – 1 mark, feature clustering (e.g., k-means) – 1 mark, use the centres of clusters to form the codebook – 1 mark.
- ii. (K). Inverted file maps each code-word to its locations in an image database. (1 mark). Then only consider database images whose bins overlap the query image (1 mark), and use measures of histogram similarity to select best from those (1 mark)
- b. Viola-Jones object detection framework provides competitive object detection in real time. It was originally designed for face detection. It involves mainly three aspects including integral image for fast feature extraction, boosting for feature selection, and attentional cascade for fast rejection of non-face windows.
- i. Four basic types of features are extracted from an image window that is under examination. Sketch the four basic types and describe how they are used in the Viola-Jones framework to extract larger set of features. [4 marks]
- ii. Usually not each of the features in a large set is useful or relevant for detecting whether a face is present in the image window. Explain how boosting is used to select relevant features. [4 marks]

Answers:

- i. (K). 2 marks for illustration of 4 basic types of features. Each of the four feature types are scaled (1 mark) and shifted (1 mark) across all possible combinations.



- ii. (K+C). Boosting is a classification scheme that works by combining weak learners into a more accurate ensemble classifier. Given reweighted data from previous stage, train all K (160,000) single-feature linear classifier (1 mark). Select the single best classifier at this stage, i.e., select the best feature among all the features (1 mark). Combine it with the other previously selected classifiers, i.e., a subset of features are selected (1 mark). Reweight the data. Learn all K classifiers again, select the best, combine, reweight. Repeat until you have T classifiers selected. (1 mark)
- c. Convolutional Neural Nets (CNNs) have had a huge impact on computer vision in recent years. Briefly describe the components of a CNN for object recognition and show how they are connected to form a complete system.

[6 marks]

Answers: (K) 1 mark for each component within the convolutional layer (i.e., Normalisation, filters, non-linear activation, feature pooling). 1 mark for classifier layer. 1 mark for the general connection between layers (i.e., the output of feature pooling is the input to the normalisation of the next convolution layer (0.5 mark), and input to the classifier is a full connected layer (0.5 mark)).

